Homework 13
ENGR072
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Problem Statement

This report aims to evaluate the optimal set up for a zipline given a set of constraints and objectives. The zipline is characterized by a combination of a set of physical parameters and optimized parameters. Three separate riders are evaluated on this zipline: a 30kg rider, a 70kg rider, and a 140kg rider. The objective of this problem is to optimize the zipline height drop and wire tension such that the maximum speed of the 70kg rider is maximized within a set of constraints.

The constant physical parameters of this problem include:

- span of the zipline = 500 m
- wire rope weight per length = 15 N/m (assume this is constant throughout the entire zipline)

The constraints of this problem are:

- vertical drop must be between 10m and 40m
- wire rope tension must be less or equal to 50 kN
- all riders must reach the end of the zipline
- all riders' exit velocities must not exceed 5m/s

Solution Approach

This solution approach can be divided into 3 parts: **(1)** evaluating the rider's path, **(2)** evaluating the rider's velocity with respect to its position on the zipline, and **(3)** optimizing the rider's velocity to maximize the maximum speed of the 70kg rider while staying within the problem's constraints. This part of the report will systematically go through each part of the solution and use graphs and equations to support the mathematical approach.

This solution references a paper written by C. Mungan and T. Lipscombe from the physics department at the U.S. Naval Academy to obtain the proper differential equations for evaluating the relationships between a rider and the path they take on a zipline.

1. Evaluating the rider's path along the zipline

This solution begins with establishing a relationship between the shape of the zipline with no rider, and the path the rider takes while travelling along the zipline. Mungan and Lipscombe define a set of equations that define the position of the rider on the zipline normalized to the entire length of the zipline. The zipline is defined on an x-y coordinate axis such that the origin is the upper leftmost corner of the zipline, and the y axis points downward.

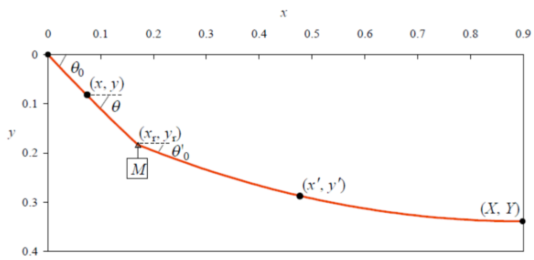

Figure 1: Shape of the zipline with the rider, where x_r and y_r indicate the rider's position, x and y indicate an arbitrary position to the left of the rider, x' and y' indicate some arbitrary position to the right of the rider. Angles θ_0, θ, and $\theta_{0'}$ are also defined on the graph. [1]

This solutions calls the MATLAB function file `rider_path.m` in order to solve for the rider's path along the zipline. This function file takes in a mass m and a tension T_0 as input, and outputs a set of x and y coordinates for the rider's path along the zipline. It is important that this file takes in only T_0 and m as the input variables in order to facilitate the optimization later on in the solution. First, the function uses Equations 1, 2, 9, and 11 from the paper to create a 3 equation system to solve for a_0, b_0, and L. The system of equations are as follows.

$$b_0 = \frac{mg}{T_0 \cos(\arctan(a_0))} \qquad \text{(Eq.1 and Eq.2)}$$

$$\tilde{X} = \frac{1}{b_0} \ln\left(\frac{a_0+\sqrt{1+a_0^2}}{a_0-b_0+\sqrt{1+(a_0-b_0)^2}}\right) \qquad \text{(Eq.9)}$$

$$\tilde{Y} = \frac{\sqrt{1+a_0^2}-\sqrt{1+(a_0-b_0)^2}}{b_0} \qquad \text{(Eq.11)}$$

This system of equation outputs a value for a_0,b_0, and L. These values are then used to solve a second system of equations, which uses combines equations 8, 16, 9 and equations 10, 17, and 11 respectively to solve for a and b parameters.

$$\frac{1}{b} \ln\left(\frac{a+\sqrt{1+a^2}}{a-bf+\sqrt{1+(a-bf)^2}}\right)+\tilde{x}'+\frac{1}{b} \ln\left(\frac{a-bf-b\gamma+\sqrt{1+(a-bf-b\gamma)^2}}{a-b\gamma-b+\sqrt{1+(a-b\gamma-b)^2}}\right)=\frac{1}{b_0} \ln\left(\frac{a_0+\sqrt{1+a_0^2}}{a_0-b_0+\sqrt{1+(a_0-b_0)^2}}\right) \qquad \text{(Eq. 8,16,9)}$$

$$\frac{\sqrt{1+a^2}-\sqrt{1+(a-bf)^2}}{b}+\frac{\sqrt{1+(a-bf-b\gamma)^2}-\sqrt{1+(a-b\gamma-b)^2}}{b}=\frac{\sqrt{1+a_0^2}-\sqrt{1+(a_0-b_0)^2}}{b_0} \qquad \text{(Eq. 10,17,11)}$$

Once a and b are solved for, the values are then inputted back equations 8 and 10 to solve for the rider's position along the zipline with respect to the leftmost edge of the zipline. into The MATLAB `rider_path.m` file then outputs a set of x and y coordinates that correspond to this pathway. Both equations 8 and 10 output the rider's position normalized to the overall length L of the zipline, so the coordinates are multiplied by L prior to output to ensure that the values correspond to the global x and y coordinate system.

2. Evaluating the rider's velocity with respect to its x position on the zipline

This solution then moves on to evaluate the rider's velocity with respect to its x position on the zipline. This is done in a new MATLAB function file `rider_velocity`, which takes inputs mass m and tension T_0 and outputs a vector of x values and a corresponding vector of $\frac{d(v^2)}{dx}$ values that correspond to the x values. This is done by calling the previous `rider_path` function to obtain a series of x and y coordinates for a given m and T_0 value, then using the MATLAB `spline` command to create a piece-wise 3rd degree polynomial of the path. Taking the derivative of this function using MATLAB's `fnder` command creates a series of $\frac{dy}{dx}$ derivatives. Using a force balance equation, an equation for $\frac{dv}{dt}$ can be obtained. [2]

$$\frac{dv}{dt} = g \sin\left(-\arctan\left(\frac{dy}{dx}\right) - F_{drag}\right)$$

Here, F_{drag} can be defined as $\frac{1}{2}\rho C_d A v^2$, where $\rho = 1.2\ kg/m^3$is the density of air, $C_d = 1$ is the coefficient of drag, $A = 0.05 m^{\frac{2}{3}} m^2$ is the frontal area of an upright rider, and v is the velocity of the rider. The derivatives $\frac{dy}{dx}$ and $\frac{dv}{dt}$ can be combined into the overall differential equation for $\frac{dv^2}{dx}$. [2]

$$\frac{dv^2}{dx} = 2\frac{dv}{dt}\sqrt{1+\left(\frac{dy}{dx}\right)^2}$$

In this differential equation, since $\frac{dy}{dx}$ is a known, and $\frac{dv}{dt}$ can be written in terms of v^2 multiplied by a series of constants, then the only independent variable of the differential equation is v^2. Using MATLAB's `ode45` function, the differential equation can be solved for v^2.

The `rider_velocity` function is called in 2 separate MATLAB function files `max_velocity` and `exit_velocity` respectively. Both of these function files take in 2 inputs, mass m and tension T_0, and output the square of the maximum velocity and the exit velocity respectively.

3. Optimizing the rider's velocity to maximize the maximum speed of the 70kg rider

This solution uses all the previously mentioned MATLAB function files to solve the overall optimization problem. The MATLAB function `fmincon` minimizes a given function, and can also take in a series of nonlinear or linear constraints to the system. In this problem, the objective is to minimize the tension of the zipline cable while maximizing the maximum speed the 70kg rider reaches, while staying within the constraints listed in the problem statement.

To make this optimization simpler, the constraints are categorized as "active" or "inactive" constraints. Inactive constraints are constraints that are implied by other constraints, and therefore do not need to be included in the optimization.

Inactive Constraints

1. The vertical drop is constrained to anywhere between 10m to 40m. However, one of the goals of this optimization is to maximize the maximum speed of the 70kg rider. Since speed on a zipline can only be increased by increasing the vertical drop, then it makes sense to assume that the vertical drop must equal the maximum 40m, and that the other constraints can be achieved by manipulating the overall tension of the zipline cable. Therefore, throughout this optimization, the vertical drop is hard-coded as 40m.
2. The second set of constraints involves ensuring that each rider's exit velocity is less or equal to 5 m/s. Given the 3 different weight riders, it makes qualitative sense to assume that if the heaviest rider is able to have an exit velocity within this constraint, then the 2 lighter riders can be assumed to have an exit velocity even less than that. Therefore, the 30kg and 70kg rider do not need to have to have an exit velocity constraint imposed on them.
3. The constraint stating that the rider's must reach the end of the zipline only applies to the lightest 30kg rider. If the 30kg rider reaches the end of the zipline, then it can be assumed that the heavier riders will definitely reach the end of the zipline, since they will accumulate more speed than the 30kg rider.

From this qualitative analysis, the constraints of this optimization are reduced to the following: **(1)** the 140kg rider must have an exit velocity less than or equal to 5 m/s, and **(2)** the 30kg rider must reach the end of the zipline. These can be written in the nonlinear constraints of the `fmincon` function. The objective function is to maximize the 70kg rider's maximum velocity, so since `fmincon` minimizes functions, the negative of the function for the velocity of the 70kg rider is inputted as the objective function. Finally, since `fmincon` is solving for tension T_0, then the lower bound is set as 0 (since the tension cannot be negative), and the upper bound is set as 50 kN, which was one of the constraints posed in the problem statement.

The overall optimization function was written in the following form.

```
constraint_max_velocity = @(T0) -max_velocity(70,T0);
lb = 0; % tension cannot be negative
ub = 50000; % upper bound for T0
function [c,ceq] = mycon(T0)
    % Redefine exit_velocity for only the 140kg rider and only the 30 kg rider.
    c(1) = exit_velocity(140,T0)-25; % subtract 5 m/s to set equal to zero
    c(2) = -exit_velocity(30,T0); % negative exit velocity of 30kg rider=
    ceq = [];
end
T0 = fmincon(constraint_max_velocity,40000,[],[],[],[],lb,ub,@mycon,options)
```

Results and Analysis

From the optimization above, the ideal tension for the cable that satisfies all the constraints is **39569 N**. This value of tension had $a = 0.1232$ and $b = 0.1263$ as the dimensionless parameters that defined the rider's path on the zipline. The overall length of the zipline was L = 502.3530. The maximum velocity reached by the 70kg rider is **12.8846** $\boldsymbol{m/s^2}$.

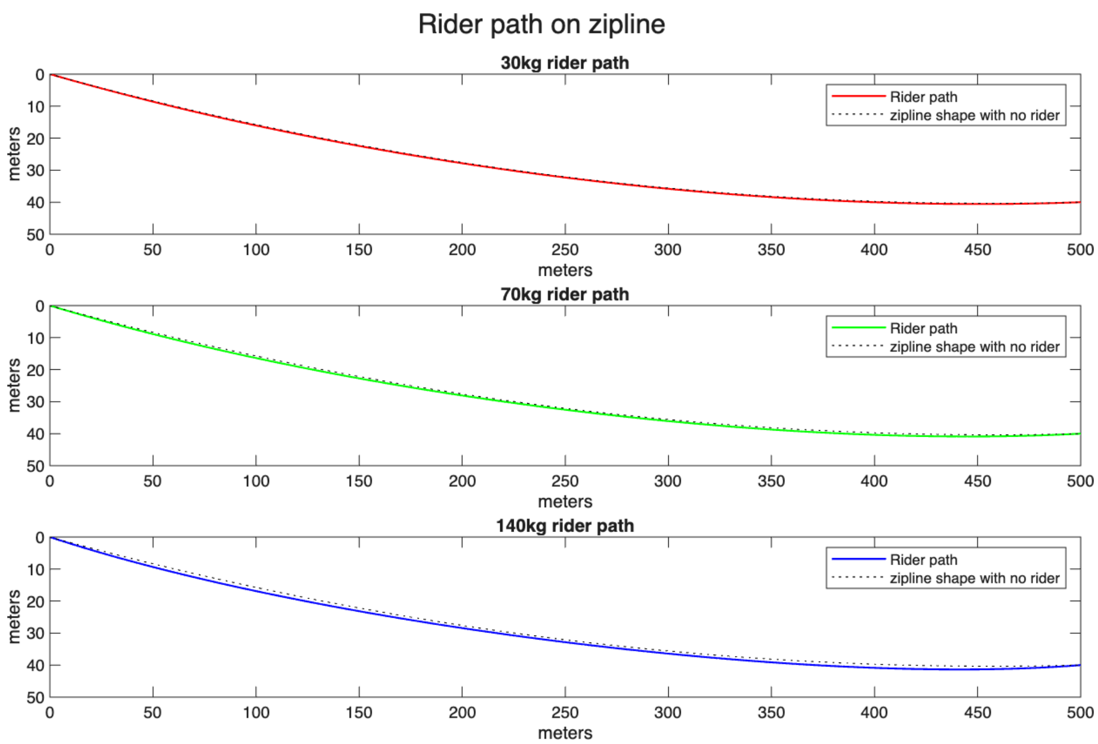

Figure 2: Each rider's path on the zipline respectively. The y axis is set such that positive y points downwards.

From Figure 2, it is clear that each rider has a vertical drop of 40 meters, which falls within our constraints. In addition, all 3 riders reach the 500 meter mark on the x axis, indicating that all riders reach the end of the zipline, satisfying the constraints. On the graph, the black dotted line indicates the shape of the zipline when there is no rider on it. It is clear that for each rider, the path is below the shape of the riderless zipline, which makes sense. In addition, looking closely, it can be noted that the 140kg rider has the largest deviation from the original zipline shape. This makes sense, since it is expected that the heaviest rider exerts the most force on the zipline, causing it to sag more when the rider is on it.

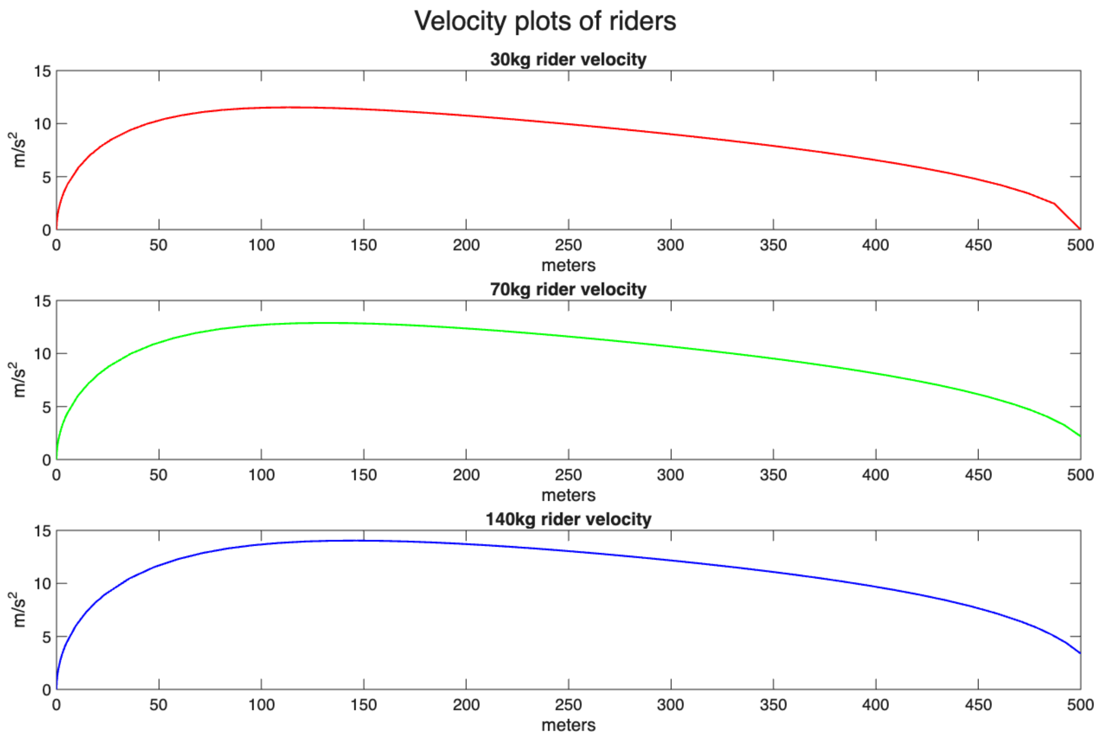

Figure 3: Each rider's velocity with respect to their x position. All velocities are positive, because in the position plot, the y axis was pointing downwards, so all riders are moving in the positive x and y direction as they travel along the zipline.

The velocity profile shows that all three riders have exit velocities less than 5, satisfying the constraint given in the problem. To be sure, the exit velocities were computed separately. The 30kg rider had an exit velocity of 0.02213 m/s^2, the 70kg rider had an exit velocity of 2.183 m/s^2, and the 140kg rider had an exit velocity of 3.3779 m/s^2.

The evaluated ideal tension value of 39569 makes sense qualitatively due to the length of the overall zipline. The length of the entire zipline is only 2.35 meters greater than the horizontal span of the zipline. As a sanity check, the hypotenuse between a horizontal span of 500 meters and vertical drop of 40- meters is 501.5974 meters, and the zipline length should definitely be larger. There is not a lot of slack in the zipline, so it makes sense that a higher tension value is needed in order to ensure that the zipline remains connected on both sides.

Earlier in the optimization, an assumption was made that the maximum height drop of 40 meters was needed to maximize the speed of the 70kg rider to reach 12.8846 m/s^2. To see if this claim holds after the optimization, using the ideal tension, the maximum speed of the 70kg rider at a vertical drop of a vertical drop of 35 meters and 30 meters was obtained. At a vertical drop of 35 meters, the maximum speed of a 70kg rider was 12.4329 m/s^2, and at a vertical drop of 30 meters, the maximum speed was 11.9716 m/s^2. This justifies the hypothesis that a 40 meter drop is needed to maximize the speed.

Reflection on ENGR072

1. One of the most interesting things I learned in ENGR072 was breaking up larger periodic functions into separate smaller trigonometric functions using the Fourier Transform Series. This was really cool because of its applications to the real world. Being able to break up complicated signals into simpler, computable signals seems almost like a secret way to look at the world through a mathematical lens. Thinking about all the noisy data and external interferences in my data from E80, I can totally see how being able to decompose the signal into simpler parts. For example, our team experienced a lot of noise in our temperature sensor data due to motor currents. I'm imagining that if I somehow knew the frequency of the signals my motors were outputting, I could use Fourier Transform to remove those signals from my data, and help me figure out what's my true measured temperature.

   Another one of the most important things I learned in E72 was MATLAB skills overall. As someone with minimal coding and computing experience, I was most nervous about entering a class that required heavy computing to solve problems. However, the way MATLAB was introduced as a part of the solution, rather than a catch all calculator helped me develop confidence using MATLAB as a tool, rather than a kind of "go-to" solution. Now, I feel confident using it to plot graphs, compute difficult optimizations, and perform other rigorous computations. In addition, I am also confident in determining when MATLAB shouldn't be used, and doing the math by hand is actually the simpler route. Building this ability to evaluate when to use a tool and when not to has built my confidence as an engineer and mathematician, and I'm really happy I got to practice this in ENGR072.

2. The advice I would give to future E72 students would be to just try the problems before immediately giving up. While this may seem obvious, I think that all the new content in E72 can be daunting, and seeing new problems could trigger a sort of "I can't do this" first instinct. However, I found that looking through the problem before articulating questions in office hours actually showed me that I knew more than I thought I did, and over time, helped me develop instincts for solving more complex problems.

3. Meetings with my mentor group helped ground me at the start of the semester. It was a little overwhelming to be in so many difficult classes at once, and so having a consistent meeting with the same people every week was really nice to reset and get a perspective of how everyone else was working through things. I think all the office hour support, mentor meetings, and TBP help in this program is awesome! One possible suggestion could be to have more than one TA at office hours for more difficult homework weeks, but I totally understand how that can be difficult to predict and manage.

Acknowledgements

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References

[1] C. Mungan and T. Lipscombe, “Traveling along a zipline,” Am. J. Phys. Educ, vol. 5, no. 1, 2011, Available: https://www.usna.edu/Users/physics/mungan/_files/documents/Publications/LAJPE6.pdf
[2] L. Bassman, Q. Yang, D. Yong, “E72 SP26 Homework 13”, homework assignment for ENGR072, Engineering Dept., Harvey Mudd, Claremont, CA, 2026. Available: https://drive.google.com/file/d/1biD154iLXZW_VJXS-jLC0ThQwvPDoOYi/view?usp=drive_link